



TRANSCENDENT
CONCUSSION RESEARCH
PROGRAM

CHEO Research Institute

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Concussion Patient Situations:

Dealing with a combination of physical, cognitive, and emotional symptoms.

- **Physical Symptoms:** Headaches (often described as pressure in the head), dizziness, loss of balance, nausea, and unusual fatigue.
 - **Sensory Issues:** Increased sensitivity to light (photophobia) and noise, blurred or double vision, and ringing in the ears (tinnitus).
 - **Cognitive Issues:** A feeling of "brain fog," difficulty concentrating, slowed thinking, and short-term memory problems (e.g., repeating the same question or forgetting events).
 - **Emotional Changes:** Unprovoked irritability, nervousness, heightened anxiety, or unexpected sadness and mood swings.
 - **Sleep Disturbances:** Sleeping significantly more or less than usual or having trouble falling and staying asleep.
 - *Note: While most symptoms resolve within 2 to 4 weeks, some patients may experience prolonged effects known as post-concussion syndrome.*
 - **Emergency Red Flags**
 - It is critical to seek immediate emergency care if the patient experiences any of the following serious warning signs:
 - Uneven pupil sizes or **dilation**
 - A severe **headache** that worsens over time
 - Repeated **vomiting**
 - **Seizures** or convulsions
 - Clear or **bloody fluid** draining from the nose or ears
 - Extreme **drowsiness** or inability to wake up
-

Concussion CRS Requirements:

Operationalizing clinical practice guidelines (CPGs) at the point of care while ensuring **safety** and **speed**

- **Verifiable Grounding:** Clinical deployment requires grounding to prevent "citation fabrication" and unsupported outputs.
- The system must use the guideline recommendations as **structured knowledge units**, where each recommendation is treated as an atomic unit with a unique identifier (ID) and a section membership.
- **Structured Input and Output:**
 - **Input:** The system requires a prompt consisting of the user query and a context block containing the guideline recommendations and their associated IDs.
 - **Output:** The system must return a response in a fixed format: a short natural-language answer followed by a list of cited recommendation IDs.
- **High Clinical Coverage (Recall):** Expert clinicians define a successful response as one that includes all "missing but required" recommendations necessary for a complete and accurate clinical answer,. The system must aim to overcome the "selection bottleneck" where retrieval might exclude necessary information,.
- **Citation Integrity:** The system must avoid "invalid citations" —citing IDs that do not exist in the actual guideline. It must also aim for high precision by avoiding "wrongly added" or non-applicable recommendations,.
- **Point-of-Care Latency:** The system is required to be "latency-conscious" to remain practical for medical environments. This involves balancing the need for complete coverage with the goal of minimizing response time (observed at approximately 25 to 45 seconds in the pilot),.
- **Context Management:** The system must be capable of processing either the **full guideline document** (to maximize recall) or a **retrieved subset** of recommendations (to minimize latency) while managing risks like "attention diffusion" in large context windows



Time Pressure & Information Overload

Time pressure & information overload prevent Clinical Practice Guideline (CPG) usage at the point of care.



LLM Challenges

Standard LLMs exhibit citation fabrication and unsupported outputs. Clinical deployment requires rigorous grounding.

The Goal: Operationalize guidelines into low-latency, strictly grounded Question Answering (QA) systems without losing accuracy.

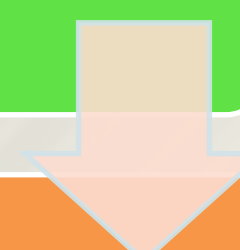


Engineering Research Questions: Guideline-grounded Clinical QA

Can *section-recommendation structures* improve precision in clinical QA?



Are there other ways to optimize the *coverage-latency tradeoff*?



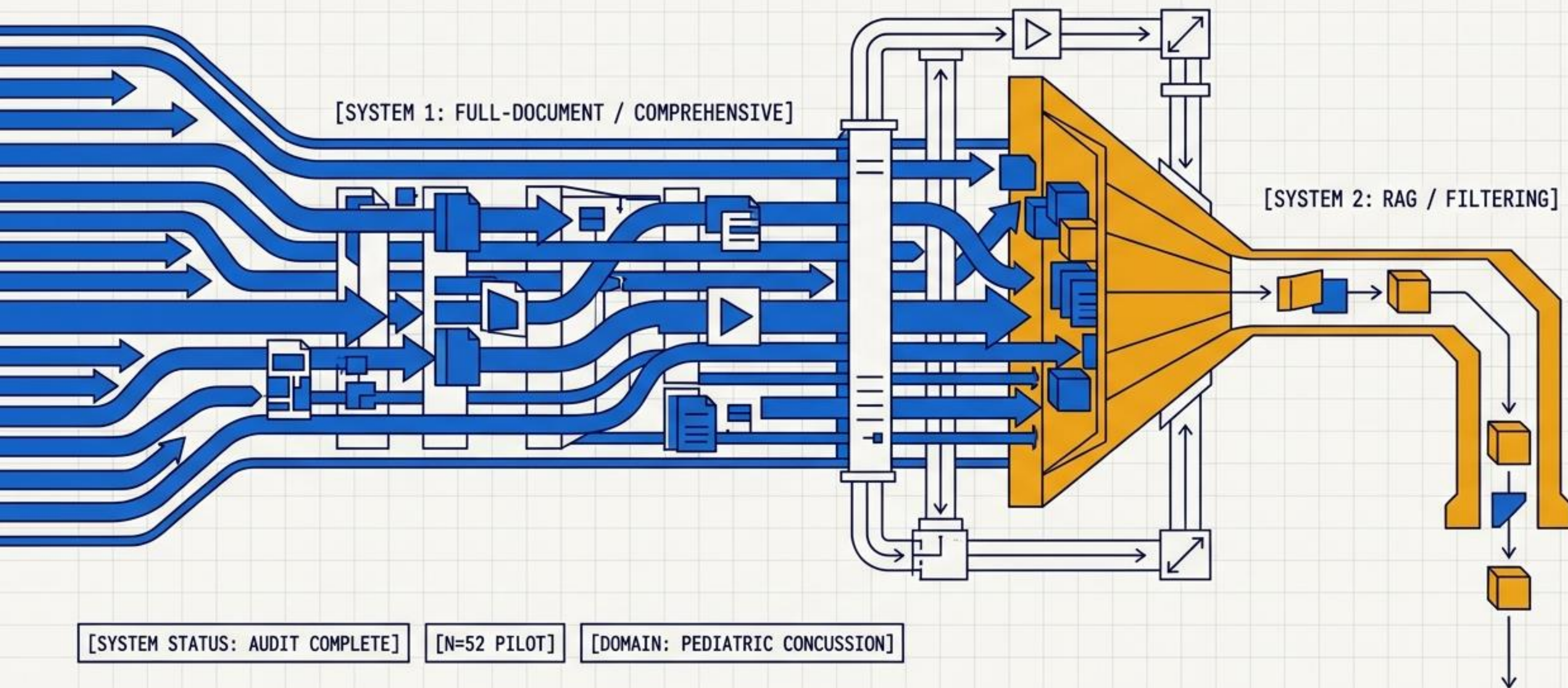
Could *adaptive retrieval strategies* improve these results?

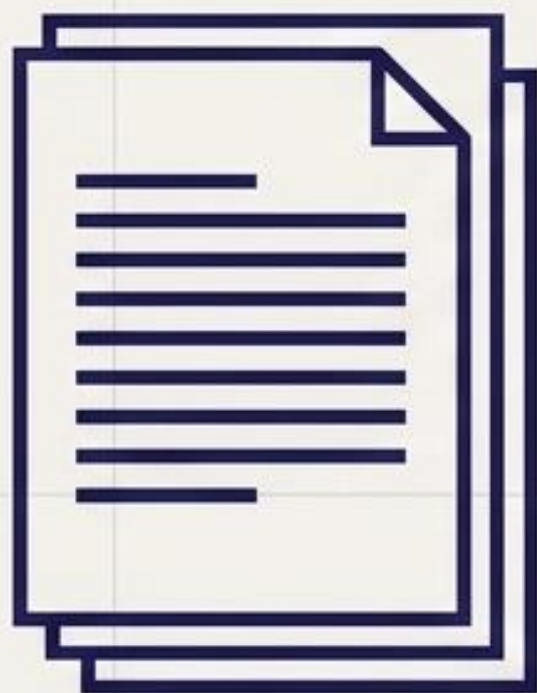




The Selection Bottleneck vs. Attention Diffusion

Quantifying Coverage–Latency Tradeoffs in Guideline–Grounded Clinical QA

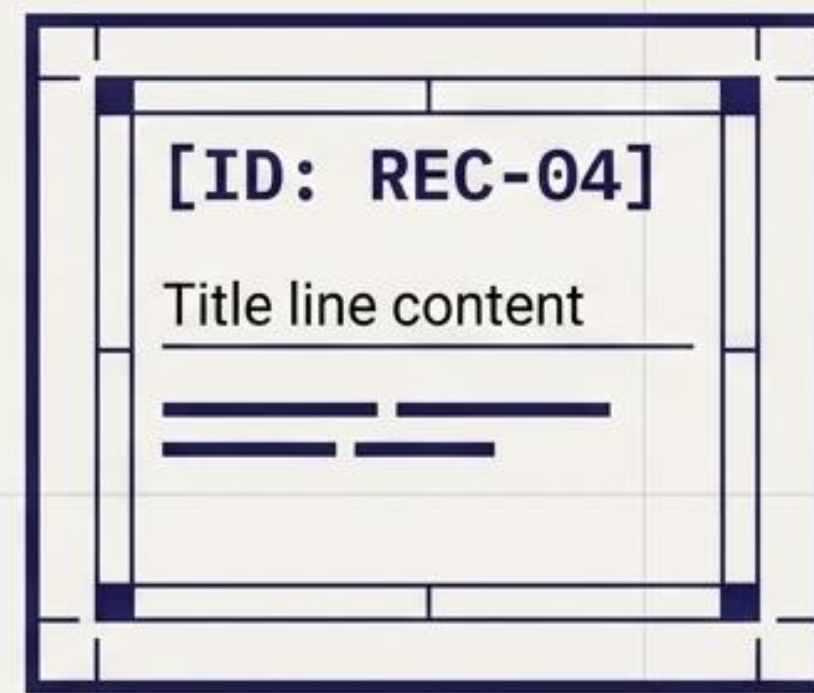




Corpus: Living Guideline for Pediatric Concussion Care



Atomic Transformation:
Recommendations as
Structured Knowledge Units



ID Assignment

Grounding requires auditing. By treating recommendations as ID-bearing atomic units, we can definitively measure exactly which guidelines an AI cites, misses, or fabricates.



The Audit Mechanism

$$G(q) = (A(q) \setminus W(q)) \cup M(q)$$

G(q): The Ground Truth
(Audit-derived reference
set for question q)

A(q): Valid IDs cited
in archived answer

W(q): IDs flagged by
clinicians as wrongly
added / not applicable

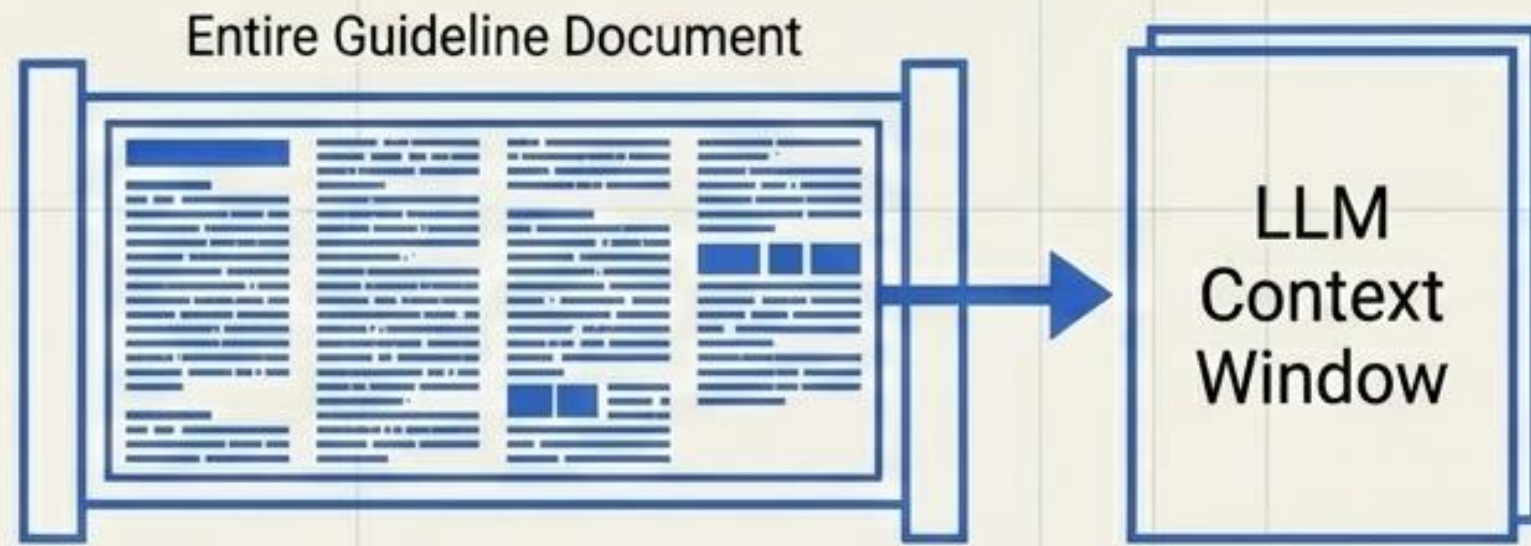
M(q): IDs flagged by
clinicians as missing
but required

Based on a retrospective audit of **N=52** clinical questions, reviewed by guideline-developing clinicians.



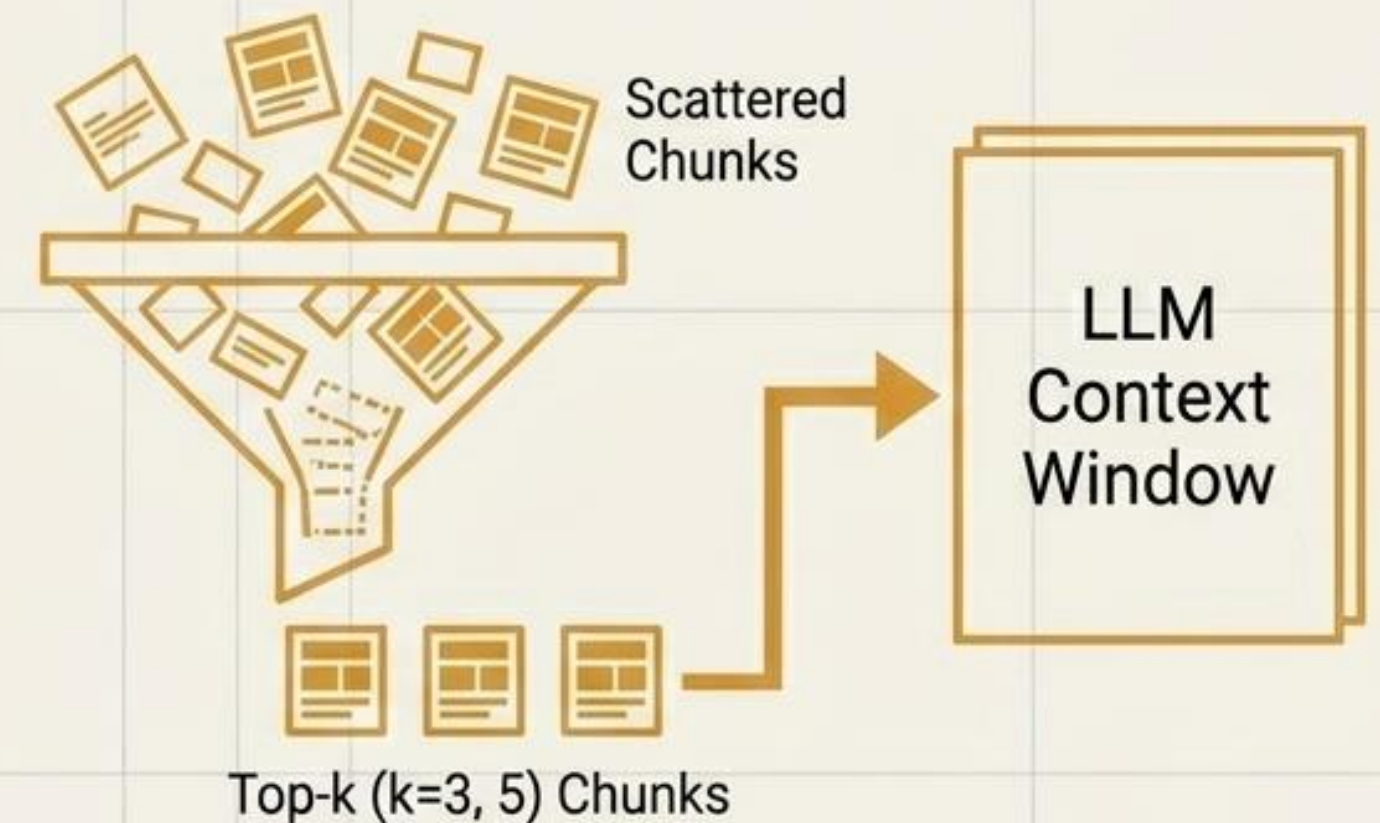
The Competing Architectures

S1: Full-Document Long-Context



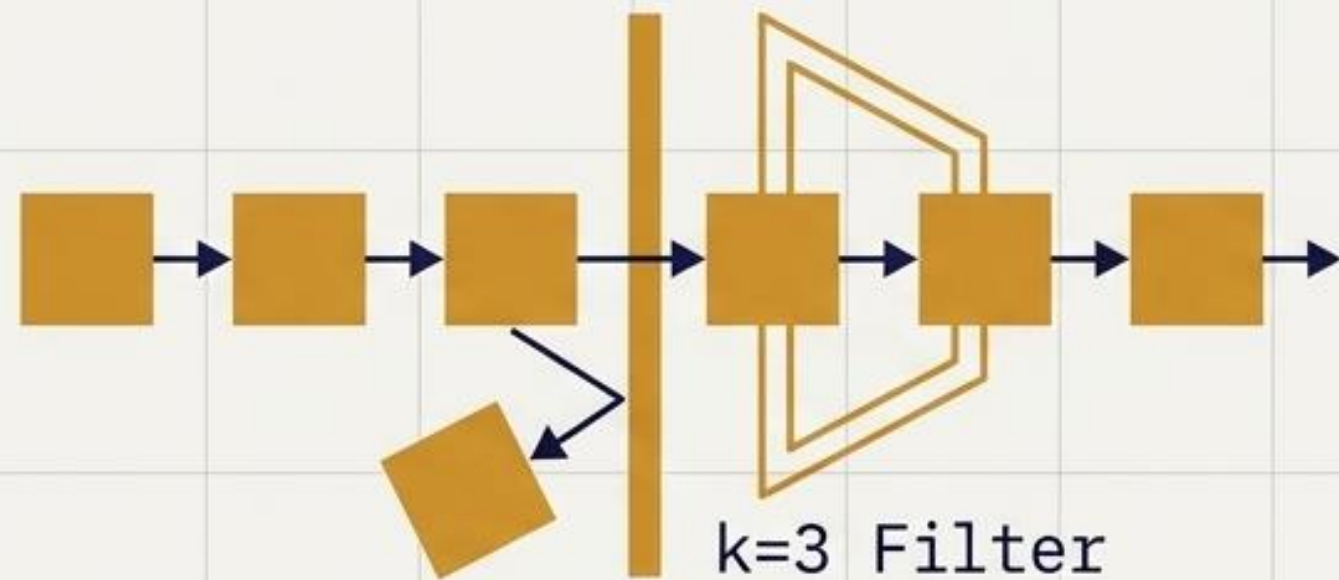
Specs: No retrieval step.
Context = Entire Guideline Document.

S2: Top-k RAG (Retrieval-Augmented Generation)



Specs: Hybrid retriever (PostgreSQL tsvector + dense text-embedding-3-large). Equal fusion. $k = \{3, 5\}$.

Risk of S2 (RAG): The Selection Bottleneck



Required recommendations are excluded by the retriever before generation even begins.

Risk of S1 (Full-Context): Attention Diffusion



The model **fails** to locate or utilize **relevant information** buried within a massive context window.



The Control Variables: Isolating the Architecture

Clinical Audit Form

- ✕ **Base Model:** `GPT-5.2` (OpenAI Responses API)
- ✕ **Reasoning Effort:** `Low` (Latency-conscious configuration for point-of-care)
- ✕ **Output Structure:** **Fixed format** (Natural language answer + cited [ID]` list)
- ✕ **Prompt Template:** Identical across systems

By holding the model and prompt constant, differences in performance strictly reflect the method of context provisioning—not model intelligence.



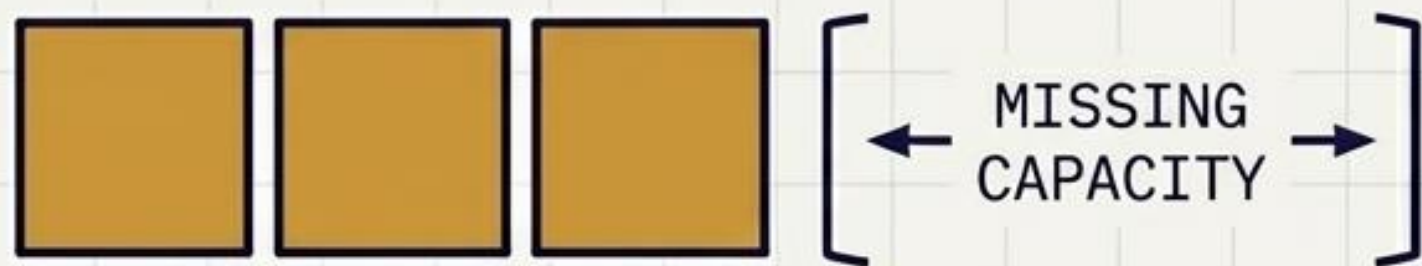
Pilot Results (N=52)

S1: Full Document		INVALID RATE	LATENCY
RECALL 1.000 	PRECISION 1.000 	0.000	 45s (Median)
S2: RAG k=3		INVALID RATE	LATENCY
RECALL 0.360 	PRECISION 1.000 	0.000	 25s (Median)
S2: RAG k=5		INVALID RATE	LATENCY
RECALL 0.473 	PRECISION 0.789 	0.000	 25s (Median)



The RAG Paradox: Trading Precision for Coverage

k=3 Context



Perfect precision (1.000), but severe **selection bottleneck**. Only 36% of required recommendations recovered.

k=5 Context



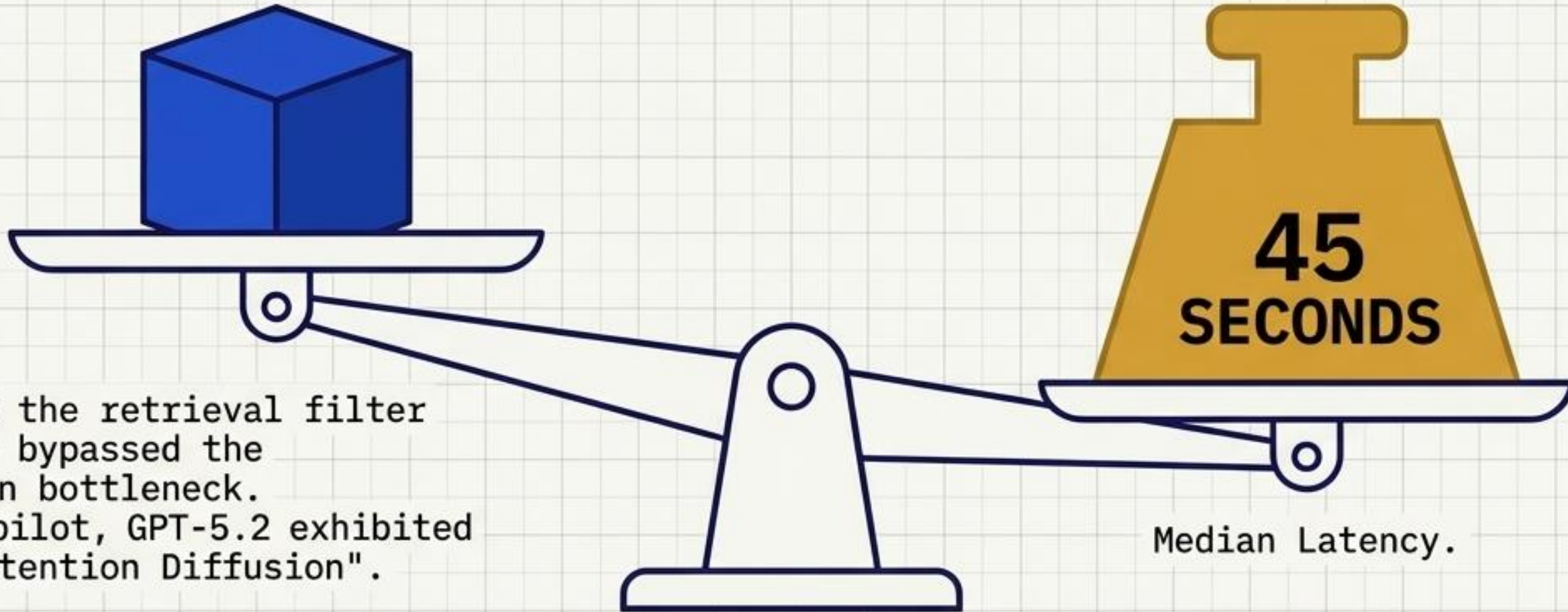
Expanding k to 5 raises **recall** (0.473) but drops precision (0.789). Additional retrieved context introduces **noise** that the model occasionally cites.

Even at k=5, recall remains < 0.5 . For many questions, the required nodes lay completely outside the top five. Re-reading retrieved context cannot recover what was never fetched.



S1 Breaks the Ceiling, but Hits a Wall

Perfect Recall &
Precision (1.000)

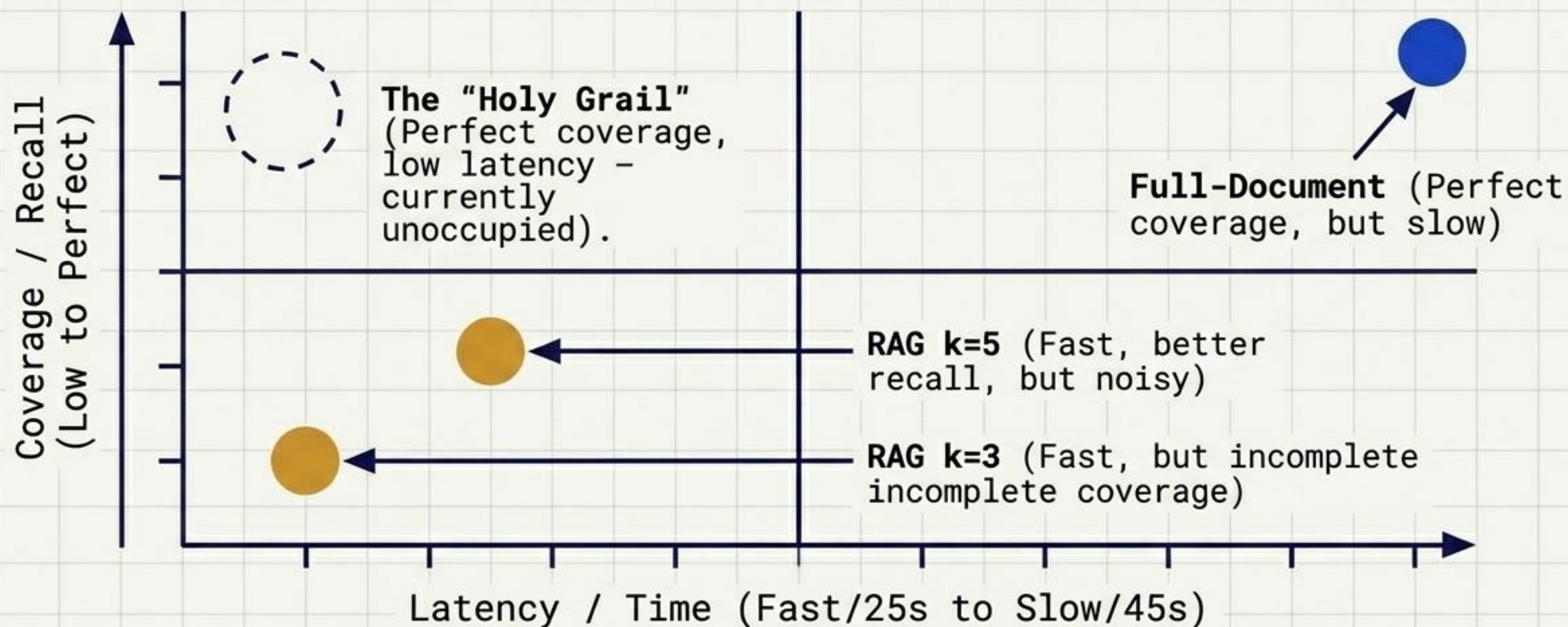


Removing the retrieval filter entirely bypassed the selection bottleneck. In this pilot, GPT-5.2 exhibited zero "Attention Diffusion".

A 45-second wait time is often unacceptable in high-urgency point-of-care environments (e.g., emergency pediatric concussion care). Perfect grounding is achieved, but clinical utility is severely compromised by speed.



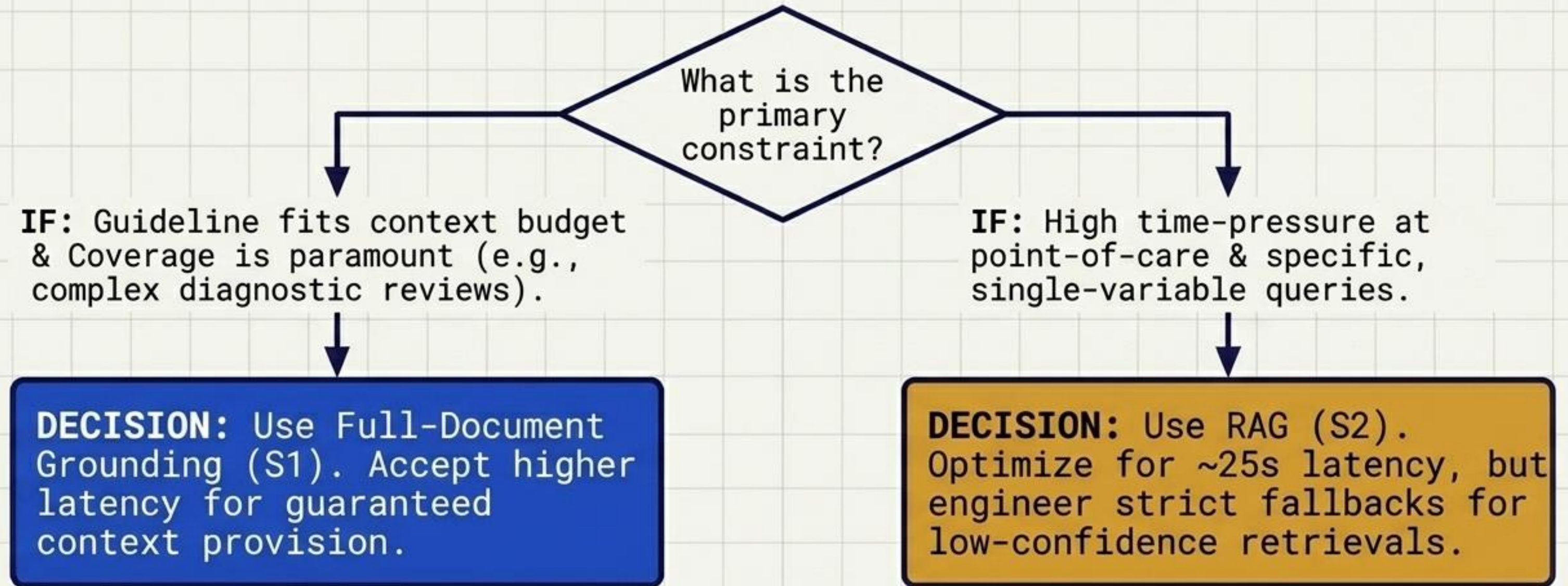
The Coverage vs. Latency Tradeoff



There is no universal 'best' method. Grounding strategy is a strict, context-dependent design choice governed by the clinical application's time constraints.



Architectural Decision Matrix



CRUCIAL NOTE: Both methods achieved a 0.000 Invalid Citation Rate. Grounding to ID-bearing atomic units successfully prevents hallucination in both architectures.



Beyond the Pilot (N=52)

Limitations

Pilot restricted to a single guideline, one model configuration, and retrospective, disjoint expert review.

The Next Frontier: Adaptive Retrieval

Moving beyond fixed top-k to dynamic retrieval algorithms that adjust context window size based on query complexity.

Scalability Analytics

Future work must decompose end-to-end latency, isolate exact token consumption costs, and expand to prospective, multi-guideline datasets.



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Research Questions: Guideline-grounded Clinical QA

Can section-recommendation structures improve precision in clinical QA?

- Yes, sections are fundamental to structured grounding and citation checking, recommendation level accuracies.

Are the other ways to optimize the coverage-latency tradeoff?

- Adaptive retrieval strategies, alternative better tuned retrievers, refined model configurations, latency decompositions, deployment economic models, guideline structures and KG.

Could adaptive retrieval strategies improve these results?

- Beyond fixed top-k is a necessary step to better understand how balance:
 - Coverage
 - Latency





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